

		[1] for correct expression
6(a)(i)	From Fig. 6.2, there is <u>constant rate of decrease of magnetic flux density</u> at right angles to the plane of the coil. By Faraday's Law, <u>there is constant rate of decrease of magnetic flux linkage in the coil</u>. Hence, the induced e.m.f. across the ends of the coil is constant from $t = 0$ s to $t = 2.5$ s. <i>Comments:</i> - many confused between magnetic flux and magnetic flux linkage, as well as the correct usage of Faraday's law. - many did not refer to the graph in their explanation, as required by the question. - use words in qualitative explanation, not mathematical symbols and equations.	[1] for correct explanation [1] for correct explanation
6(a)(ii)	By Faraday's law, e.m.f. = rate of change of magnetic flux linkage $\langle E \rangle = \frac{\Delta\Phi}{\Delta t} = \frac{\Delta(NBA)}{\Delta t}$ $\langle I \rangle R = NA \frac{\Delta B}{\Delta t}$ $\langle I \rangle \Delta t = \frac{NA\Delta B}{R}$ $\Delta Q = \frac{NA\Delta B}{R} \left(\because I = \frac{\Delta Q}{\Delta t} \right)$ $= \frac{180 \times \frac{\pi}{4} (6.4 \times 10^{-2})^2 \times 1.4 \times 10^{-2}}{1.5}$ $= 5.405 \times 10^{-3}$ $\approx 5.4 \times 10^{-3} \text{ C}$ <i>Comments:</i> - many did not read the scale on the vertical axis correctly, omitting the order 10^2 in their calculation. - many used the formula $B = \mu_0 n I$ without understanding the question fully.	[1] for $\langle E \rangle = \frac{\Delta\Phi}{\Delta t}$ $= \frac{\Delta(NBA)}{\Delta t}$ [1] for correct expression and numerical substitution [1] for correct answer

	- many mistakenly took the rate of decrease of the magnetic flux in the solenoid as the induced e.m.f. in the coil C.	
6(b)(i)	Using $P = IV$, $I_s = \frac{P}{V} = \frac{15}{6.0} = 2.5 \text{ A}$ By $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ and $V_s I_s = V_p I_p$, $I_p = \frac{N_s}{N_p} I_s$ $= \frac{25}{1000} \times 2.5$ $= 0.0625$ $= 6.25 \times 10^{-2} \text{ A}$	[2]
6(b)(ii)	When the battery is first connected, <u>current increases instantaneously in the primary coil, giving rise to increase in magnetic flux linkage in secondary coil</u>. Hence, there is induced e.m.f. across the lamp. After some time, the <u>current in the primary coil is constant</u>. There is <u>no more change in magnetic flux density in secondary coil</u>. Hence, no e.m.f. across lamp. <i>Comments:</i> - many explanations were incomplete concentrating their response to the p.d. across the lamp being zero after some time. - the question requires the application of Faraday's law from the moment the battery is connected.	[1] for correct explanation [1] for correct explanation
7(a)(i)	13.6 eV	[1] for correct